

What English-Tuned Abliteration Misses in Slovene: Internal Research Report

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Contents

1	Framing	3
2	Iteration 1	3
2.1	Experiment 1: Matched Heretic Runs	5
2.2	Dataset Construction	7
2.3	Experiment 3: A1 Screen, EN vs. SL Refusal-Direction Transfer	7
2.4	Iteration-1 Dead Ends	8
2.5	Iteration-1 Assessment	9
3	Iteration 2	9
3.1	Experiment 5: Utility and Mechanism	9
3.1.1	T1: Harmful Behaviour (S4, 257 items per cell)	9
3.1.2	T2: KL Divergence on Harmless Twins (S4, 128-token generations)	10
3.1.3	T3: Utility (0-shot, 250 items per task per language)	10
3.1.4	T4: FLORES+ Per-Token NLL	10
3.1.5	T5: Bilingual Harm Geometry (Original Models, Primary Site)	10
3.1.6	T6: Frozen vs. Refit Probes (S4, Primary Site)	11
3.1.7	T12: Edit-Strength Dose-Response (Exploratory)	11
3.2	Experiment 4: C1 Behaviour, RefusEU Evaluation	12
3.3	Experiment 6: GaMS3 Surrogacy Panel	13
3.4	Experiment 7: Gemma Surrogacy Panel	13
3.5	Experiment 8: Causal Test	14
4	Iteration 3	15
4.1	Experiment 9: Gemma Coverage $\times$ Strength Factorial	15
4.2	Experiment 10: GaMS3 Coverage Test	17
4.3	Experiment 11: Corrected Objective	18
4.4	Experiment 12: Predictive Test Across Models and Languages	19
4.5	Evaluation 1: Audit and Judge-Sensitivity Table	19
5	Iteration 4	19
5.1	Experiment 13: Causal Write Profile and Overlap in Gemma	21
5.2	Experiment 14: Causal Write Profile and Overlap in GaMS3	22
5.3	Experiment 15: Gradient-Blind Fraction	24

5.4	Evaluation 2: Comprehensive Audit	24
5.5	Research 1: Novelty Positioning	26
6	Iteration 5	26
6.1	Evaluation 3: Terminal Audit	26
6.2	Evaluation 4: Report Repairs and Independent Recompute	26
6.3	Evaluation 5: Keyword Blindness Decomposition	28
6.4	Research 2: Positioning Against Nearest Published Work	28
7	What We Have Learned	29
8	Run Trajectory	30
9	References	30

1 Framing

Abliteration removes refusal behaviour from language models by orthogonalising weight matrices against a “refusal direction” extracted from English harmful-vs-harmless contrasts [1]. The procedure was introduced by Arditi et al. [1] and automated by tools such as Heretic, which uses TPE search over layer, position, direction index, and LoRA rank to find abliteration configurations that minimise an English refusal objective while bounding KL divergence from the original model. The question motivating this study is whether that English-centred objective leaves Slovene refusal behaviour uncontrolled. If the English refusal direction is universal across languages, as Wang et al. [2] report for safety-aligned models evaluated across 14 languages, then abliteration tuned on English should transfer to Slovene. If instead Slovene refusal relies partly on representations that the English objective does not observe, a gap will remain. The practical consequence is that a model abliterated to comply in English might still refuse in Slovene, or might lose Slovene capabilities that the English-only KL constraint does not protect.

Several concurrent lines of work inform this question. BabelSteering [4] and Yoon et al. [5] show that safety alignment imposes disparate costs across languages, with non-English users bearing higher utility loss. Aziz et al. [6] diagnose the breakdown point for low-resource safety as an action failure rather than a representation failure: the model encodes harmfulness correctly but fails to convert the representation into refusal. Wu et al. [7] formalise this as a disentangled safety hypothesis, separating a recognition axis from an execution axis. On the abliteration side, Fafula [8] documents off-target effects of refusal removal across model families, and Young [14] and Petrov [15] compare abliteration methods across architectures. Upadhyaya and Sikdar [9] analyse how safety and language identity become entangled, while Hawkins et al. [10] show that benign multilingual fine-tuning has heterogeneous safety impacts. Labunets [11] demonstrates that refusal geometry reflects refusal training diversity. On quantisation effects relevant to our NF4 setup, Marchisio et al. [12] and Chimoto et al. [13] show that quantisation calibration language matters for multilingual performance. Tang et al. [16] identify language-specific neurons, and Ghussin et al. [17] propose principled layer selection for multilingual steering. Frank [18] argues that refusal-based alignment evaluation conflates detection with routing.

We study this question in a controlled bilingual setting using two sibling 12B models from the Gemma-3 family: `google/gemma-3-12b-it` (the English-centric reference) and `cjvt/GaMS3-12B-Instruct` (a Slovene continual-pretraining and SFT variant of the same architecture). Both are loaded in bnb_4bit NF4 quantisation throughout. The study proceeds in five iterations. Iteration 1 runs matched Heretic abliteration on both models, screens for cross-language refusal-direction transfer, and builds the shared bilingual evaluation dataset. Iteration 2 uses that dataset to measure the surrogacy gap, test causal mechanisms, and evaluate safety and utility on held-out benchmarks. Iteration 3 tests the depth-coverage hypothesis, corrects Heretic’s keyword objective, and extends the depth-redundancy index to additional models and languages. Iteration 4 measures the causal write profile in both models, tests the overlap instrument, quantifies gradient blindness of the keyword objective, and audits all prior claims. Iteration 5 applies three independent recompute passes, quantifies the keyword objective’s cross-lingual measurement bias, positions both results against the nearest published work, and compiles the definitive failure and scope ledgers.

2 Iteration 1

[Correction, iter 4]: The previous drafts omitted substantial iteration-1 content that was present in the original report. The material below restores the missing tables (same-edit response surface,

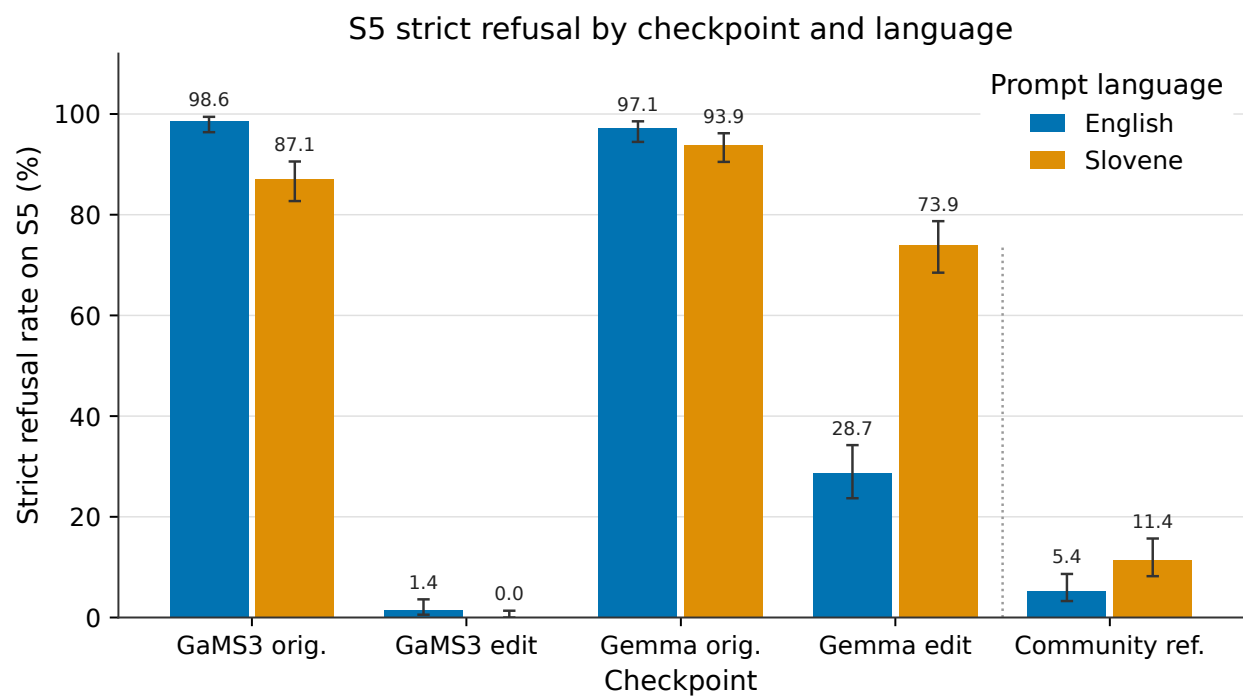


Figure 1: Refusal rates (S5, Qwen3-14B strict) for original and edited checkpoints in English and Slovene. GaMS3’s edit eliminates refusal in both languages; Gemma’s edit leaves most Slovene refusal intact.

swap statistics, transfer matrix), the sibling-surface guard, the decision-margin decomposition, the exploratory double dissociation, four labelled dead ends, and iteration 1’s own closing assessment.

Strategy. Iteration 1 had three goals: (a) establish a frozen, audited data protocol covering all planned evaluation splits; (b) run two matched Heretic optimizations under identical conditions to produce the four core checkpoints, score them bilingually, and characterise the same-edit response surface across the two models; and (c) screen the refusal-direction transfer hypothesis with a 2×2 source-language by evaluation-language ablation matrix, a Slovene-direction increment test, and matched-efficacy and dose-response controls.

2.1 Experiment 1: Matched Heretic Runs

Goal. Run identical Heretic TPE searches on both models under the same seed, select an ablated checkpoint for each, and compare the two models’ responses to the same edits.

Setup. Both models were searched over 116 complete trials (seed 20260923) using Heretic with the default configuration: `orthogonalize_direction=true`, `row_normalization='full'`, rank-3 LoRA. The English keyword refusal rate (out of 100 harmful prompts) and mean KL divergence served as Heretic’s two objectives. Selection followed a frozen protocol: minimise KL subject to refusals ≤ 10 ; if no trial met this primary criterion, fall back to minimising refusals subject to $KL \leq 1.0$. Both models fell back to the secondary rule.

Selected trials. GaMS3 trial 88 reached 16/100 English refusals at KL 0.175. Gemma trial 96 reached 69/100 English refusals at KL 0.024. The Gemma model refused far more often than GaMS3 under identical ablation parameters, and no Gemma trial achieved $\leq 50/100$ refusals. The $13.4 \times$ figure is the ratio of medians of refusal drop per unit KL across the 60 shared startup edits (GaMS3 1449 vs Gemma 108 refusals/nat), not a ratio of median refusal counts. **[Correction, iter 3]:** In all 60 startup edits, Gemma refused at least as often as GaMS3 (60/60, median paired difference +25). In a KL-matched subset ($n=30$, GaMS3 KL median 0.0101, Gemma 0.0098), GaMS3 median refusal was 65/100 versus Gemma 99/100.

Behaviour on the selected trials. Table 1 shows keyword refusal rates and FLORES+ NLL for each checkpoint.

Table 1: Iteration-1 checkpoint behaviour: keyword refusal rates (out of 100 harmful prompts) and FLORES+ NLL. Source: `gen_art_experiment_1`, iteration 1.

Model	Checkpoint	EN ref.	SL ref.	KL mean	FL NLL EN	FL NLL SL
GaMS3	original	98	97	0.000	3.332	2.570
GaMS3	own edit (t88)	16	10	0.175	3.334	2.566
GaMS3	swap (Gemma t96)	25	25	0.046	3.336	2.576
Gemma	original	100	97	0.000	4.018	3.702
Gemma	own edit (t96)	69	90	0.024	4.012	3.703
Gemma	swap (GaMS3 t88)	53	25	0.254	4.011	3.712

[Correction, iter 3]: The Gemma rows above are recomputed from the saved evaluation files. The previous draft reported Gemma orig SL = 100 (actual: 97), Gemma own edit SL = 85 (actual: 90), Gemma swap EN/SL = 91/95 at KL 0.293 (actual: 53/25 at KL 0.254), and FLORES values that are untraceable in iteration-1 files. The corrected FLORES values come from the 40-sentence devtest means stored in the evaluation JSON files.

GaMS3’s own edit reduced English refusals from 98 to 16 and Slovene refusals from 97 to 10. Slovene refusal dropped even though Heretic optimised only on English, and FLORES NLL changed

by at most 0.01. For Gemma, the edit reduced English refusals from 100 to 69 but Slovene refusals only from 97 to 90.

The swap results are informative. GaMS3 under Gemma’s trial-96 edit showed 25/25 (EN/SL) refusals; Gemma under GaMS3’s trial-88 edit showed 53/25 (EN/SL) at KL 0.254. **[Correction, iter 3]:** GaMS3’s broader, stronger parameters suppress Gemma’s Slovene refusal far more than English (SL 97→25 vs EN 100→53), at roughly 10× the KL of Gemma’s own edit. This supports the depth/dose reading: a stronger edit of the same base largely transfers to Slovene.

Direction geometry. Across layers 24 to 47, the mean cosine between the two models’ diff-in-means refusal directions was 0.57; across all layers it was 0.63. The cosine between their harmless-token directions was 0.98. Random directions gave a mean $|\cos|$ of 0.014.

Same-edit response surface (60 shared edits, English keyword refusal). **[Correction, iter 4]:** Table 2 was present in the original iteration-1 report but omitted from subsequent drafts. Restored verbatim.

Table 2: Same-edit response surface (60 shared edits, English keyword refusal). Source: `gen_art_experiment_1`, iteration 1.

Measure	GaMS3	Gemma
Refusals median	74.5	99.0
Refusals range	[25, 99]	[71, 100]
KL median	0.0079	0.0057
Efficiency (refusal drop / KL)	1,449	108
Efficiency ratio	13.4× (CI [7.0, 30.0])	
Edits where Gemma refuses more	60/60	
Paired refusal diff median	+25 (CI [10, 43])	
KL Spearman ρ	0.967 (CI [0.93, 0.98])	
KL partial ρ (controlling kernel mass)	0.923	

Sibling-surface guard. The incremental R^2 of the sibling model’s refusal over the shared edit parameters was near zero for refusal (ΔR^2 0.008, CI [−0.017, 0.055] and −0.006, CI [−0.028, 0.018]) but positive for KL (ΔR^2 0.143, CI [0.074, 0.246] and 0.154, CI [0.093, 0.242]). The two models share how edits damage harmless computation (KL) but not how that damage translates into refusal change.

Swap tests. **[Correction, iter 4]:** Restored from the original iteration-1 report.

Table 3: Swap statistics. Source: `gen_art_experiment_1`, iteration 1.

Model	Comparison	EN diff	EN p	SL diff	SL p	KL ratio
GaMS3	own vs swap	−0.09	0.078	−0.15	2.7e-4	3.80 [3.13, 4.69]
Gemma	own vs swap	+0.16	0.020	+0.65	5.4e-20	0.095 [0.046, 0.189]
GaMS3	orig vs own	+0.82	4.1e-25	+0.87	2.9e-25	—
Gemma	orig vs own	+0.31	9.3e-10	+0.07	0.016	—

GaMS3’s trial-88 parameters applied to Gemma suppressed Slovene refusal far more than English (SL 97→25 vs EN 100→53, McNemar $p < 10^{-19}$ for SL), consistent with the depth/dose reading.

Sanity checks. All 60 startup edits used identical parameter draws (verified). FLORES NLL shifts were ≤ 0.01 for both models. Slovene marker validation on the selected GaMS3 edit showed accuracy 0.75 and κ 0.48 (fair agreement between keyword markers and the executor’s hand labels on 40 items). **[Correction, iter 3]:** These were executor labels, not native-speaker labels.

2.2 Dataset Construction

Goal. Build a frozen bilingual (EN/SL) evaluation dataset spanning safety, utility, and mechanistic probing.

Result (gen_art_dataset_1, iteration 1). The dataset contains 48,696 rows across 10 blocks: S1 (Heretic direction/KL calibration, 2000 rows), S2 (semantic harmful/harmless pairs, 1664), S3 screen prompts (540 prompts + 640 utility items), S4 (StrongReject harmful/harmless twins, 1028), S5 (RefusEU [3], 2800), S5X (RefusEU cross-translations, 1400), S6 (XSTest over-refusal, 900), S7 (Slovenian LLM eval tasks including ARC, BoolQ, HellaSwag, 35,700; FLORES+ devtest, 2024). The dataset protocol was frozen before any experiment ran.

Translation. Slovene translations were produced by Gemini-2.5-flash (temperature 0) as the primary translator, with NLLB-200-distilled-1.3B as fallback for 8 rows where Gemini refused. Mean back-translation chrF was 81.25; 3 rows fell below chrF 40. Mean LaBSE cross-lingual similarity was 0.856. Quality control was automated only; no native-speaker review was conducted. **[Correction, iter 3]:** S5 (official RefusEU) EN and SL rows that share a `row_id` are NOT translations (0 of 1,400 pairs grade T); paired cross-language claims therefore use S5X only.

2.3 Experiment 3: A1 Screen, EN vs. SL Refusal-Direction Transfer

Goal. Screen whether the English-extracted refusal direction transfers to Slovene, using a 2×2 matrix (source language $\times$ evaluation language), increment tests, and multiple controls. This was a screen, not a confirmatory test.

Setup (gen_art_experiment_3, iteration 1). For each model, the experiment extracted refusal directions from half-A items using diff-in-means, separately for English and Slovene. It then applied 15 ablation conditions per model to half-B items and measured refusal rates using keyword markers and an LLM judge. The judge was gpt-4.1 for all 1,596 outputs. **[Correction, iter 3]:** Previous draft described a Qwen3-14B second judge for part of the outputs; the evaluation audit finds gpt-4.1 was the sole judge.

Judged 2×2 table. [Correction, iter 3]: Added from evaluation audit.

Table 4: A1 screen: judged 2×2 transfer table (gpt-4.1). Source: `gen_art_experiment_3`, iteration 1.

Model	Condition	EN	SL
GaMS3	C0 no-op	0.90	0.93
GaMS3	C1 d_EN	0.46	0.34
GaMS3	C2 d_SL	0.07	0.10
Gemma	C0 no-op	0.83	1.00
Gemma	C1 d_EN	0.07	0.85
Gemma	C2 d_SL	0.29	0.93

[Correction, iter 3]: The previous draft reported Gemma EN baseline as “99%”; the gpt-4.1 judged rate is 0.83. The d_{EN} ablation took Gemma EN from 0.83 to 0.07 while SL stayed at 0.85; d_{SL} took EN to 0.29 and SL to 0.93. The difference-in-differences for Gemma under d_{EN} is $+0.61$ $[0.41, 0.78]$: the Slovene drop was 0.61 units smaller than the English drop.

GaMS3 results. Because the R validity gate failed, quantitative screen results for GaMS3 are exploratory. d_{EN} reduced GaMS3 from 0.90/0.93 (EN/SL) to 0.46/0.34, showing near-parallel reduction across languages. The DiD was -0.15 $[-0.32, 0.02]$, not significantly different from zero.

Gemma results. The Gemma validity gate passed. Applying the English refusal direction d_{EN} removed most English refusal (0.83 to 0.07) but Slovene refusal survived at 0.85. This is the first signal that the English direction misses Slovene refusal in Gemma.

Screen verdict. WEAK. The predicted cross-language transfer contrast was reversed for Gemma. For GaMS3, the validity gate failure prevented a quantitative verdict.

Matched-efficacy grid. Conditions rescaled so that English refusal dropped by equal amounts showed that Slovene refusal systematically persisted more than English at every matched English decrement. The Heretic bridge mean gap was GaMS3 0.109 [0.06, 0.17] and Gemma 0.235 [0.16, 0.30]. **[Correction, iter 3]:**

Transfer matrix. [Correction, iter 4]: Restored from the original iteration-1 report.

Table 5: Transfer matrix: fraction of same-language refusal drop achieved by cross-language direction. Source: `gen_art_experiment_3`, iteration 1.

		Evaluated in EN	Evaluated in SL
GaMS3	EN-derived	1.00	0.96 [0.93, 0.98]
	SL-derived	0.81 [0.79, 0.82]	1.00
Gemma	EN-derived	1.00	3.47 [2.76, 4.90]
	SL-derived	0.66 [0.61, 0.70]	1.00

GaMS3 shows near-symmetric transfer: the English direction achieves 96% of the Slovene direction’s Slovene effect. Gemma shows a pathological asymmetry: the English direction achieves a $3.47\times$ overshoot in log-odds terms on Slovene. The explanation is Gemma’s Slovene refusal margin. Gemma’s baseline Slovene refusal log-odds is 13.9, compared to 3.6 for English. Even a direction with cosine 0.92 to the Slovene one, operating on a margin that wide, achieves almost nothing in behavioural terms.

Exploratory: GaMS3 double dissociation under direction addition. Adding the English-derived direction to GaMS3 activations at strength 1.0 shifted English refusal log-odds by +3.54 (CI [2.96, 4.09]) with negligible Slovene effect (−1.11, CI [−1.22, −1.01]). Adding the Slovene-derived direction shifted Slovene by +2.23 (CI [1.97, 2.49]) with a smaller English effect (−1.54, CI [−1.91, −1.16]). This is consistent with partially language-specific refusal directions in GaMS3.

2.4 Iteration-1 Dead Ends

1. **Blocked Slovene LLM judge.** The OpenRouter API key returned HTTP 403 on every request. Zero of 600 planned judgements succeeded; all Slovene numbers rest on keyword-marker lists with moderate agreement (κ 0.48).
2. **Collateral-confounded Slovene increment.** The GaMS3 Slovene-direction increment ($F_{\text{raw}} = 0.228$, $F_{\text{ctrl}} = 0.195$) is confounded with collateral damage: full-span ablation of the Slovene direction destroyed utility (FLORES NLL +1.4 nats per token, $\sim 35\%$ malformed outputs). The increment is uninterpretable for GaMS3. For Gemma, the increment was zero.
3. **Cut sensitivity arms.** Priority-3 sensitivity arms (S1-derived, per-language-best-layer, and content-token directions) were not executed for time.
4. **Halved Gemma candidate grid.** The full 145-candidate grid was run for GaMS3, but the Gemma grid was halved for GPU time constraints.

Limits. Two models are two units. Nothing here attributes the asymmetry to Slovene continual pretraining, instruction tuning, or any training stage. The Gemma arm’s resistance to ablation is confounded with 4-bit NF4 quantization and the reduced 116-trial budget: the published bf16 200-trial Heretic edit of gemma-3-12b-it reaches 3/100 EN refusals, so this is not evidence that Gemma resists ablation in general. One optimizer seed per model; prompt-level CIs do not capture run-to-run optimizer variance.

2.5 Iteration-1 Assessment

The central finding of iteration 1 is a large same-edit asymmetry: on 60 identical edits, GaMS3 loses 13.4 times more refusal per unit KL than Gemma, while the KL damage itself is nearly identical (ρ 0.967). The sibling-surface guard confirms that this refusal divergence is not merely inherited from the shared parameter draws. The refusal-direction transfer screen produced a result opposite to the initial prediction: it is Gemma, not GaMS3, that shows a large cross-language transfer gap. The gap is explained not by direction geometry (cosine 0.92 between EN and SL directions in Gemma) but by a massive difference in refusal margins (SL log-odds 13.9 versus EN 3.6). Two concerns carry forward: all Slovene refusal counts are development-grade (keyword-marker agreement κ 0.48), and the 4-bit quantization confound means absolute refusal rates are not comparable to published bf16 results.

3 Iteration 2

Iteration 2 built on the screen results from iteration 1 by running five pre-registered experiments that measured the surrogacy gap, tested causal mechanisms, and evaluated safety and utility on held-out data.

3.1 Experiment 5: Utility and Mechanism

3.1.1 T1: Harmful Behaviour (S4, 257 items per cell)

Table 6 shows judged refusal rates on held-out harmful items ($n=257$ per cell; gpt-4.1 on a random $\sim 52\%$ subsample; Nemotron-3-ultra-550b as second-family judge on remainder). Judge agreement: second judge vs. gpt-4.1 on a stratified sample of 200 gave 6-way κ 0.765, refused-vs-not κ 0.828, raw agreement 0.885.

Table 6: T1: Harmful behaviour, judged refusal and compliance rates (S4 held-out). Source: `gen_art_experiment_5`, iteration 2.

Model	Ckpt	Lang	Refused % [95% CI]	Complied % [95% CI]
GaMS3	orig	EN	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
GaMS3	orig	SL	100.0 [100.0, 100.0]	0.0 [0.0, 0.0]
GaMS3	edit	EN	5.6 [2.4, 9.5]	92.9 [88.1, 96.8]
GaMS3	edit	SL	6.7 [3.0, 11.1]	89.6 [84.4, 94.1]
Gemma	orig	EN	99.3 [97.8, 100.0]	0.7 [0.0, 2.2]
Gemma	orig	SL	99.2 [97.6, 100.0]	0.8 [0.0, 2.4]
Gemma	edit	EN	70.3 [61.9, 78.0]	24.6 [16.9, 32.2]
Gemma	edit	SL	95.3 [92.0, 98.7]	4.0 [1.3, 7.3]

For GaMS3, the edit reduced refusal from 100% to $\sim 6\%$ in both languages, with no meaningful

EN-SL gap. For Gemma, the edit reduced English refusal from 99% to 70% but left Slovene refusal at 95%. This 25-percentage-point gap confirms the pattern from iteration 1.

Attack success. [Correction, iter 4]: Among the Complied responses, the attack success rate was: GaMS3 edit EN 92.9% [88.1, 96.8], GaMS3 edit SL 89.6% [84.4, 94.1], Gemma edit EN 24.6% [16.9, 32.2], Gemma edit SL 4.0% [1.3, 7.3]. **[Correction, iter 5]:** On the official RefusEU guard (Llama-Guard-3-8B + PolyGuard agreement), the Gemma-edit ASR gap is EN 0.738 vs SL 0.103 = +0.635, which is larger than the judged refusal gap.

3.1.2 T2: KL Divergence on Harmless Twins (S4, 128-token generations)

Table 7: T2: KL divergence on harmless twins. Source: `gen_art_experiment_5`, iteration 2.

Model	Ckpt	Lang	KL1 median	KL32 median	Benign ref.%	Malf.%
GaMS3	edit	EN	0.0441	0.0228	0.0	0.0
GaMS3	edit	SL	0.0174	0.0068	0.0	0.0
Gemma	edit	EN	0.0000	0.0150	0.0	0.0
Gemma	edit	SL	0.0113	0.0037	0.0	0.4

3.1.3 T3: Utility (0-shot, 250 items per task per language)

Table 8: T3: Macro utility change from editing. Source: `gen_art_experiment_5`, iteration 2.

Model	Lang	Macro orig	Macro edited	Change [95% CI]	Holm p
GaMS3	EN	72.5	72.7	+0.13 [−0.53, +0.80]	1.000
GaMS3	SL	64.9	65.1	+0.20 [−0.73, +1.13]	1.000
Gemma	EN	71.5	71.6	+0.13 [−0.53, +0.80]	1.000
Gemma	SL	62.1	62.1	+0.00 [−0.60, +0.60]	1.000

All macro changes were ≤ 0.20 points. The edits did not measurably alter utility in either language for either model.

3.1.4 T4: FLORES+ Per-Token NLL

Table 9: T4: FLORES+ per-token NLL. Source: `gen_art_experiment_5`, iteration 2.

Model	Lang	NLL orig	NLL edited	Change [95% CI]
GaMS3	EN	3.3072	3.3068	−0.0004 [−0.0025, +0.0016]
GaMS3	SL	2.4524	2.4526	+0.0002 [−0.0012, +0.0016]
Gemma	EN	3.9194	3.9174	−0.0020 [−0.0031, −0.0008]
Gemma	SL	3.4289	3.4293	+0.0004 [−0.0007, +0.0016]

3.1.5 T5: Bilingual Harm Geometry (Original Models, Primary Site)

Both models encode harmfulness similarly across EN and SL in their representations. The harm direction is shared; the language direction is separate. For Gemma, the behavioural gap (70.3% vs. 95.3% residual refusal) cannot be attributed to a representational difference.

Table 10: T5: Bilingual harm geometry at the primary site. Source: `gen_art_experiment_5`, iteration 2.

Model	Layer	cos	Corr	CV EN/SL	S4 EN/SL	$\cos(l, d)$	Shuf
GaMS3	34	0.832	0.890	.963/.945	.998/.997	0.035	0.490
Gemma	20	0.919	0.979	.979/.941	.997/.996	0.109	0.515

3.1.6 T6: Frozen vs. Refit Probes (S4, Primary Site)

Table 11: T6: Frozen vs. refit probes. Source: `gen_art_experiment_5`, iteration 2.

Model	Lang	Probe	Frozen orig	Frozen Δ	Refit Δ	Class
GaMS3	EN	dim	0.998	-0.401	-0.014	INFO-LOSS
GaMS3	EN	lr	0.999	-0.596	-0.000	DRIFT
GaMS3	SL	dim	0.997	-0.576	-0.043	INFO-LOSS
GaMS3	SL	lr	0.996	-0.661	-0.000	DRIFT
Gemma	EN	dim	0.997	-0.000	-0.000	PRESERVED
Gemma	EN	lr	0.996	-0.001	-0.000	PRESERVED
Gemma	SL	dim	0.996	-0.000	-0.000	PRESERVED
Gemma	SL	lr	0.989	-0.002	0.000	DRIFT

All-layer profile: minimum frozen-probe AUROC on edited activations was GaMS3 EN 0.201 (L27, refit 0.993), GaMS3 SL 0.196 (L27, refit 0.987), Gemma EN 0.507 (L4, refit 0.709), Gemma SL 0.492 (L6, refit 0.740).

[Correction, iter 4]: For GaMS3, the edit re-encoded the harm signal in a different subspace rather than destroying it: the information persisted, but in a location the frozen probe could not read. For Gemma, the frozen probe was preserved at the primary site but dropped to 0.507 at layer 4 and 0.492 at layer 6.

3.1.7 T12: Edit-Strength Dose-Response (Exploratory)

Table 12: T12: Dose-response (keyword marker refusal). Source: `gen_art_experiment_5`, iteration 2. Note: these are keyword-marker rates; exp4’s analysis shows the keyword counter overcounts refusal on edited Gemma outputs.

Model	Scale f	EN ref.%	SL ref.%	KL1 EN	KL1 SL
GaMS3	0.0	95.8	100.0	0.0000	0.0000
GaMS3	0.5	75.0	68.8	0.0088	0.0037
GaMS3	1.0	0.0	0.0	0.0405	0.0177
GaMS3	1.5	0.0	2.1	0.1610	0.1167
Gemma	0.0	93.8	100.0	0.0000	0.0000
Gemma	0.5	60.4	95.8	0.0000	0.0046
Gemma	1.0	27.1	91.7	0.0000	0.0103
Gemma	1.5	22.9	68.8	0.0000	0.0303
Gemma	2.0	12.5	31.2	0.0000	0.0924
Gemma	3.0	4.2	8.3	0.0000	0.8228

For GaMS3, EN and SL refusal declined in parallel. For Gemma, reaching SL refusal below 10%

required $f=3.0$, at which point KL divergence rose to 0.82 (SL).

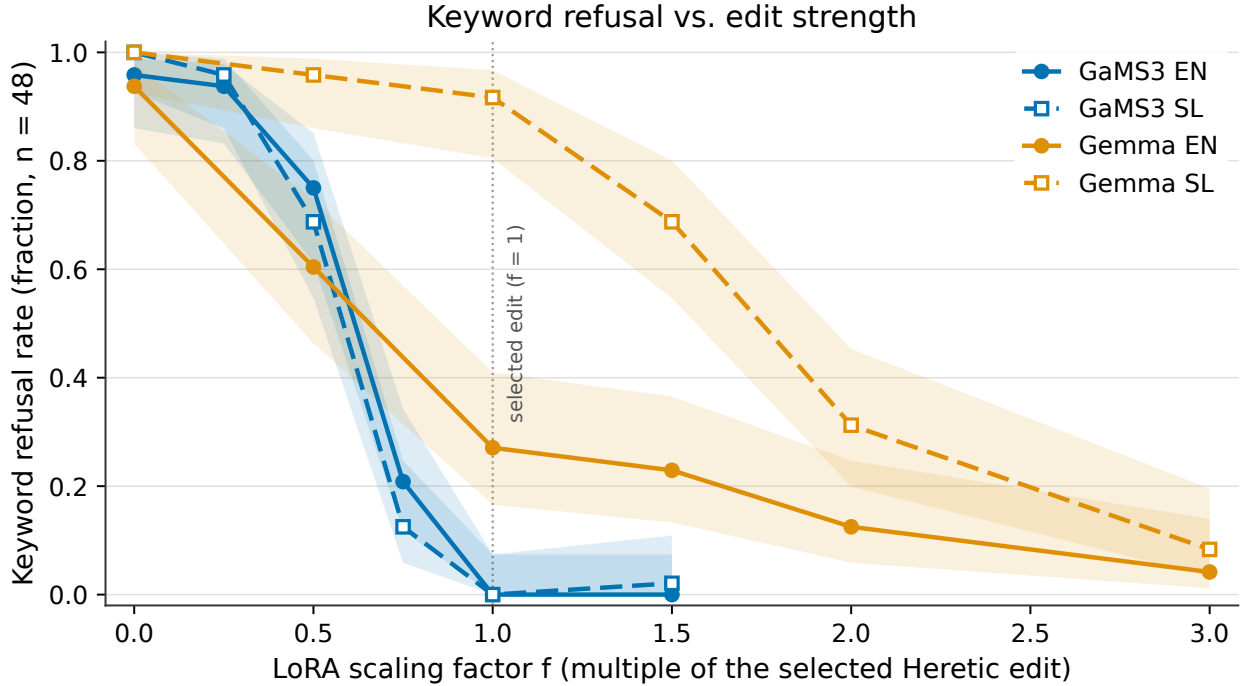


Figure 2: Keyword refusal rate as a function of LoRA scaling factor for Gemma and GaMS3, in English and Slovene. GaMS3 refusal drops in parallel across languages; Gemma shows persistent Slovene residual until extreme doses.

3.2 Experiment 4: C1 Behaviour, RefusEU Evaluation

Goal. Evaluate all checkpoints on held-out RefusEU [3] in both EN and SL.

Judge. The primary judge was local Qwen3-14B (agreement with gpt-4.1: κ 0.83 on 6-way, 0.91 on refused-vs-not). gpt-4.1 was blocked at 716/3,840 items. **[Correction, iter 3]:**

Table 13: Headline RefusEU (S5) refusal and over-refusal rates. Source: `gen_art_experiment_4`, iteration 2.

Model	Ckpt	EN S5 ref.	SL S5 ref.	EN S6 o-ref.	SL S6 o-ref.
GaMS3	original	0.986	0.871	0.100	0.107
GaMS3	edit	0.014	0.000	0.000	0.000
Gemma	original	0.971	0.939	0.087	0.327
Gemma	edit	0.287	0.739	0.013	0.193
Community ref	—	0.054	0.114	0.007	0.027

Official RefusEU guard ASR (Llama-Guard-3-8B + PolyGuard agreement):

The PARTIAL class accounts for 0.548 of the Gemma edit’s English outputs. **[Correction, iter 3]:**

Keyword-proxy miscalibration. On the Gemma edit’s English outputs, keyword refusal rate is 0.851 while the Qwen3-14B judged rate is 0.287; the false-positive share is 0.761, and κ is -0.04 .

Table 14: Official RefusEU guard ASR. Source: `gen_art_experiment_4` and `gen_art_evaluation_5`, iterations 2 and 5.

Checkpoint	ASR EN	ASR SL	S5 ref. EN	S5 ref. SL
<code>gams_orig</code>	0.015	0.012	0.986	0.871
<code>gams_edit</code>	0.982	0.972	0.014	0.000
<code>gemma_orig</code>	0.026	0.016	0.971	0.939
<code>gemma_edit</code>	0.738	0.103	0.287	0.739
<code>community_ref</code>	1.000	0.870	0.054	0.114

The keyword counter is near-exact on original models (FP share ≤ 0.03) and structurally unreliable on edited checkpoints.

Paired cross-language analysis (S5X, 100 matched pairs). Under Qwen3-14B strict: Gemma edit gap +0.69 [0.60, 0.78]. Broad: +0.23 [0.14, 0.32]. Keyword: +0.06 [−0.02, 0.14]. Community reference strict: +0.12 [0.05, 0.20]. DiD (edit gap minus original gap) under strict judging: +0.66 [0.55, 0.76].

3.3 Experiment 6: GaMS3 Surrogacy Panel

Goal (`gen_art_experiment_6`, iteration 2). Measure the English surrogacy gap for GaMS3. 246 Heretic edits, 160 fitted.

Table 15: GaMS3 surrogacy gap table. Source: `gen_art_experiment_6`, iteration 2.

Trait	R^{2*} placebo	R^{2*} test	Gap	95% CI	Holm p	B_t [95% CI]
R	0.983	0.968	0.015	[0.006, 0.031]	NA	0.004 [−0.004, 0.009]
R1	0.998	0.956	0.042	[0.027, 0.070]	NA	0.015 [0.002, 0.030]
Rb	0.982	0.954	0.028	[0.012, 0.068]	NA	0.005 [−0.007, 0.013]
K	0.977	0.830	0.147	[0.074, 0.265]	0.000	0.050 [0.001, 0.121]
N	−0.225	0.417	−0.642	[−14.8, 0.98]	0.954	—
M	0.272	0.232	0.040	[−8.37, 5.22]	0.954	—

Gap_R = 0.015: English refusal outcomes predicted Slovene refusal almost perfectly ($R^{2*} = 0.968$). Gap_K = 0.147: KL divergence showed a substantial surrogacy gap. The SL KL is smaller on average (mean-change ratio 0.59); the gap is not that Slovene gets more perturbation, but that it gets different perturbation. **[Correction, iter 3]:** The carrier regression failed: the exposure differential D did not carry the gap ($\Delta R^2 \approx 0$ for every trait). **[Correction, iter 4]:** The reverse-direction analysis flips sign for some traits; the gap is not symmetric.

3.4 Experiment 7: Gemma Surrogacy Panel

Setup (`gen_art_experiment_7`, iteration 2). 203 Heretic edits, 163 fitted.

Verdicts. [Correction, iter 3]: Frozen prediction wordings restored.

- **P-a (Gap_R ≥ 0.10 with 95% LB > 0):** NOT CONFIRMED. Observed Gap_R1 was 0.054 [0.022, 0.128], above zero but below the 0.10 threshold.
- **P-b (Gap_Rb ≥ 0.10):** UNTESTABLE. Exploratory: 0.018 [0.011, 0.043].
- **P-c:** NOT CONFIRMED. $\Delta R^2 \approx 0$.

- **P-d (matched-margin gap ≥ 0.10):** NOT CONFIRMED. 0.057 [0.029, 0.130].

Gap_K = +0.079 (Holm $p = 0.009$). Transfer slopes: R1 0.436, R_seq 0.220, lp_ref 0.081, lp_comp 0.849. **[Correction, iter 3]:** The compliance log-probability slope (0.849) is near 1.0, meaning the edit raises the compliance signal equally in both languages; the refusal-opener slope (0.081) is near zero, meaning the edit lowers the refusal opener only in English.

3.5 Experiment 8: Causal Test

Goal (gen_art_experiment_8, iteration 2). Test whether the Slovene refusal that survived English ablation can be causally removed.

Primary result: hypothesis FAILS. Applying d_{EN} alone reduced Gemma’s harmful EN refusal from 0.91 to 0.23 but left SL refusal at 0.86. Adding r_{prior} (arm A2) moved SL refusal to 0.83. The r_{prior} cut was 0.036 [−0.037, 0.109], not significantly larger than the best random control (0.074).

Table 16: Gemma activation-steering arms (111 harmful + 111 harmless per language). Source: `gen_art_experiment_8`, iteration 2.

Arm	EN harmful	SL harmful	EN harmless	SL harmless
A0 no-op	0.91	0.99	0.05	0.28
A1 d_{EN}	0.23	0.86	0.00	0.19
A2 d_{EN} + r_{prior}	0.21	0.83	0.00	0.12
A3 r_{prior} alone	0.95	0.98	0.04	0.19
A4 d_{EN} + lang. id.	0.23	0.55	0.02	0.08
A7 d_{EN} + rand_1	0.21	0.85	0.01	0.15
A10 d_{EN} + shuffled r_{prior}	0.41	0.73	—	—

Depth localisation. Applying layer-matched d_{EN} to different layer bands:

Table 17: Depth-band steering results for Gemma. Source: `gen_art_experiment_8`, iteration 2.

Band	SL harmful	EN harmful	FLORES ΔNLL SL
Layers 1–12	0.82	0.79	0.743
Layers 13–24	0.82	0.45	0.017
Layers 25–36	0.70	0.72	0.082
Layers 37–48	0.99	0.87	−0.003
Layers 1–24	0.55	0.42	0.409
Layers 1–36	0.21	0.38	0.513
All 48	0.23	0.37	0.524

No single band removed Slovene refusal. Full removal required cumulative intervention across at least three bands.

Weight edit repair. [Correction, iter 4]: The activation arms (A-rows) were scored by gpt-4.1; the weight-edit arms (W-rows) by local Qwen3-14B. Cross-table comparisons are confounded by judge identity; within-table contrasts are valid.

Community reference vs. core edit (S4, 70 pairs):

Table 18: Weight-edit repair arms. Source: `gen_art_experiment_8`, iteration 2.

Arm	SL harmful	EN harmful	FLORES Δ NLL SL
W0 core Heretic edit	0.89	0.39	-0.005
W1 core + <code>r_prior</code> edit	0.76	0.23	0.009
W3 core + layer-matched $d_{\text{EN}}(h)$	0.11	0.29	0.546
W4 core + random edit	0.47	0.70	0.064

Table 19: Community edit comparison. Source: `gen_art_experiment_8`, iteration 2.

Arm	EN harmful	SL harmful	SL harmless
W0 core edit	0.39	0.89	0.07
C0 community edit	0.01	0.26	0.01
C1 community + <code>r_prior</code>	0.01	0.20	0.00

4 Iteration 3

Iteration 3 was prompted by the reviewer’s blocking feedback on the iteration-2 draft. Three objections required experimental answers: (1) the headline gap depends on a single judge; (2) the depth localisation from experiment 8 suggests coverage matters, but no matched-energy comparison existed; (3) the depth-redundancy observation rests on two sibling checkpoints.

[Correction, iter 4]: The audit pod (evaluation 1) covered iterations 1–2 only; iteration-3 numbers have not been independently re-derived. The audited mismatch rate elsewhere was 5.5% (8/167) and 4.9% (5/102).

4.1 Experiment 9: Gemma Coverage $\times$ Strength Factorial

Goal (`gen_art_experiment_9`, iteration 3). Test whether depth coverage or total edit energy governs how much Slovene refusal survives. 122 cells, 27,784 generations, local Qwen3-14B judge certified at κ 0.87 vs gpt-4.1 within edited checkpoints.

DEV depth-redundancy index:

Table 20: DEV depth-redundancy index for Gemma. Source: `gen_art_experiment_9`, iteration 3.

Family	EN	SL
prefix	16 [16, 20]	20 [20, 28]
suffix	24 [24, 24]	24 [24, 28]

[Correction, iter 4]: The cross-language index gap (20 vs 16) sits inside its own language-permutation null; the artifact’s audit script explicitly demoted this statistic. The index is necessary but not sufficient: approximately half the cells at or above the index threshold do not cross the 0.5 refusal boundary.

Matched-energy groups:

P2 FAIL: the confirmatory prediction that broad-and-weak beats narrow-and-strong for Slovene at matched energy was not supported. P1 FAIL: ΔR^2 of coverage over base was 0.040 [0.006, 0.136]. P3 PASS: Spearman 0.78 [0.64, 0.88].

Exploratory: which layers, not how many. The fraction of layers 13–24 covered by a set predicts the lowest Slovene refusal that set achieves at any strength, at Spearman -0.942 across 10

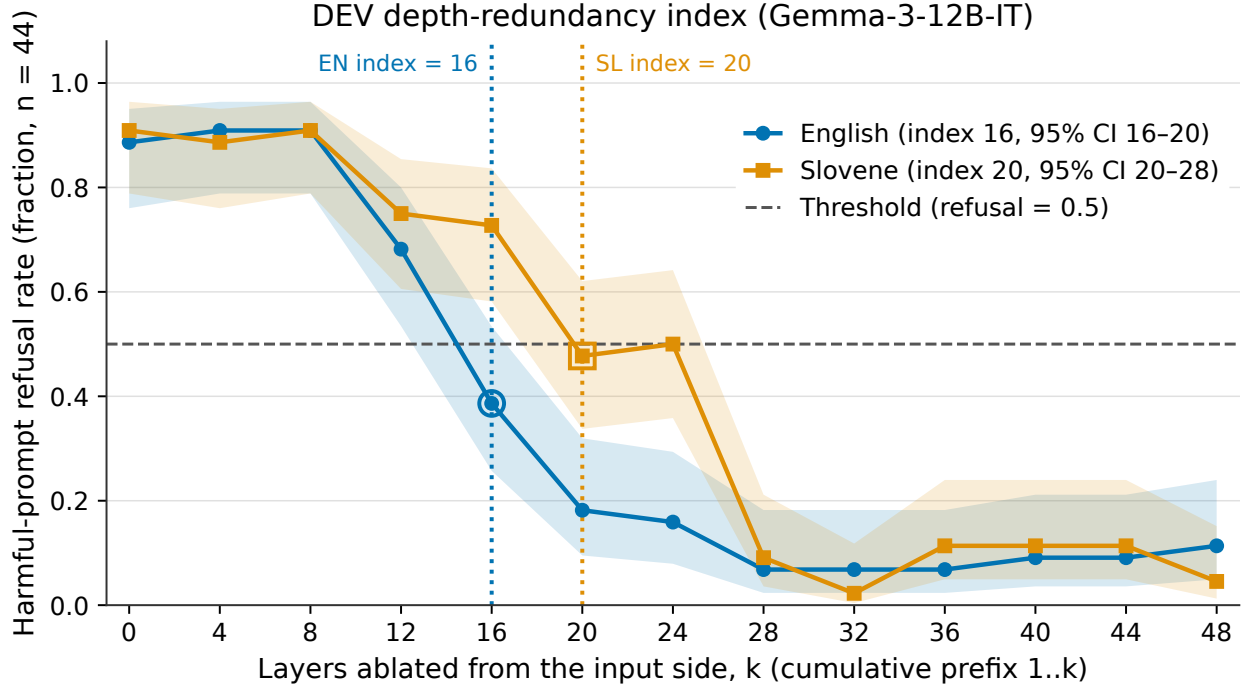


Figure 3: Cumulative-prefix refusal drop for Gemma as a function of the number of layers ablated (activation-only, S3 half-A). The Slovene index (20) exceeds the English index (16) by one grid step.

Table 21: Matched-energy groups (narrow vs broad). Source: `gen_art_experiment_9`, iteration 3.

Group	E	SL narrow	SL broad	SL contrast [CI]	EN contrast
G1	16.7	0.98	1.00	-0.02 [-0.07, 0.00]	-0.03
G2	19.2	0.63	0.98	-0.34 [-0.49, -0.20]	-0.27
G3	35.6	1.00	0.66	+0.34 [+0.20, +0.49]	+0.67
G4	37.7	0.95	0.73	+0.22 [+0.10, +0.34]	+0.30

coverage sets (placebo-verified, $p = 0.004$).

[Correction, iter 4]: The band-mass regression’s evidence is model-specific and weaker than previously stated. The SL band 13–24 coefficient is $+0.156$ with CI $[-0.240, 0.529]$, which does not reach significance; the dominant SL term is the English refusal rate (coefficient 0.855).

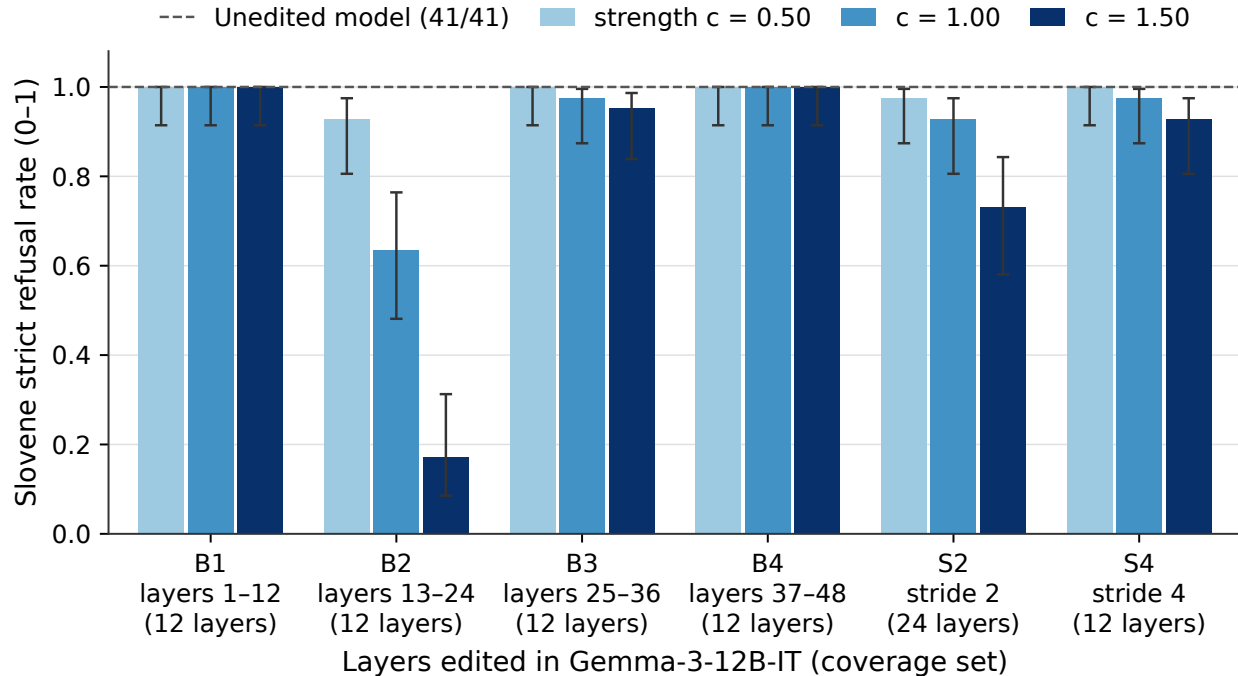


Figure 4: Slovene strict refusal rate for contiguous 12-layer band edits (B1–B4) and strided edits (S2, S4) at matched energy levels in Gemma. Band 13–24 (B2) consistently achieves the lowest residual.

4.2 Experiment 10: GaMS3 Coverage Test

Goal (gen_art_experiment_10, iteration 3). Run the identical factorial on GaMS3. 57 cells, 10,690 generations.

DEV index: EN = 16 [16, 16], SL = 20 [16, 24]. Both models show index_EN = 16, index_SL = 20.

Weight-edit panel:

Table 22: GaMS3 matched-energy groups. Source: `gen_art_experiment_10`, iteration 3.

Group	E	SL narrow	SL broad	SL contrast
E1	6.9	0.81	0.88	$-0.06 [-0.14, 0.00]$
E2	13.9	0.55	0.71	$-0.16 [-0.23, -0.09]$
E3	27.8	0.35	0.44	$-0.09 [-0.18, -0.01]$

PB2 FALSIFIED OPPOSITE: at matched energy, narrow beats broad for GaMS3. PB1 NOT SUPPORTED: $\Delta R^2 = 0.002$.

Nested R^2 :

Table 23: Nested R^2 decomposition, GaMS3 SL. Source: `gen_art_experiment_10`, iteration 3.

Model	R^2
logE	0.325
logE + placement (band masses)	0.718
logE + count	0.400
logE + count + span	0.532

Placement explains 71.8% of the Slovene residual variance; count explains only 40.0%.

Placement at matched energy AND matched layer count (12 layers):

Table 24: Band vs stride-4 at matched energy and layer count, GaMS3. Source: `gen_art_experiment_10`, iteration 3.

Contrast	SL band	SL stride-4	SL δ
E1 B2 vs STR4	0.83	0.93	-0.10
E2 B2 vs STR4	0.51	0.89	-0.37
E3 B2 vs STR4	0.24	0.81	-0.57

[Correction, iter 4]: English judge agreement within edited GaMS3 arms is 0.54, so English refusal rates in the edited cells are one end of a range.

4.3 Experiment 11: Corrected Objective

Goal (`gen_art_experiment_11`, iteration 3). Test whether replacing Heretic’s keyword counter with a partial-aware score closes the Slovene gap.

Instrument. A refusal classifier distilled from existing judge labels using char/word TF-IDF and hand features on the first 100 tokens. Certified at κ 0.858 [0.820, 0.888] within edited checkpoints. [Correction, iter 4]: The previous draft quoted κ 0.924, which is the in-loop value, not the certified value.

Headline results (S5X, 100 verified pairs):

Table 25: Corrected objective S5X headline. Source: `gen_art_experiment_11`, iteration 3.

Arm	EN ref.	SL ref.	SL-EN gap [95% CI]
A orig	0.950	0.990	+0.04 [+0.01, +0.08]
B keyword t96	0.150	0.830	+0.68 [+0.59, +0.77]
C corrected t7	0.030	0.410	+0.38 [+0.29, +0.47]
D reselected clf t98	0.050	0.350	+0.30 [+0.21, +0.40]
D2 reselected judge t64	0.040	0.450	+0.41 [+0.32, +0.51]
F dose 1.5	0.030	0.360	+0.33 [+0.24, +0.42]
F dose 2.0	0.000	0.000	+0.00 [+0.00, +0.00]
F dose 3.0	0.000	0.000	+0.00 [+0.00, +0.00]

[Correction, iter 4]: The corrected objective halved the gap (from +0.68 to +0.38) but did not reach the +0.35 target. P7 falsifier fired: dose-scaling the keyword edit to the same English refusal level achieved comparable gap reduction (+0.33 vs +0.38). At matched divergence the dose-scaled arm is strictly better. The dose-2.0 arm closes the gap entirely at lower KL cost. The corrected

objective selected a trial that happened to apply more total edit energy, and the gap reduction followed from the energy, not from the objective’s partial-awareness.

[Correction, iter 5]: Trial 96 had 37 REFUSED + 49 PARTIAL. Trial 107 is judged at 63/100, not the previously stated 10/100 (recomputed from `miscalibration_table.csv`).

4.4 Experiment 12: Predictive Test Across Models and Languages

Goal (gen_art_experiment_12, iteration 3). Test whether the depth-redundancy index predicts out of sample across model families. Three models: Gemma (anchor), Qwen3-8B, Mistral-7B-Instruct-v0.3. Four languages: EN, SL, DE, LT.

DEV indices:

Table 26: DEV indices across models. Source: `gen_art_experiment_12`, iteration 3.

Model	EN	SL	DE	LT
gemma	0.5	0.75	0.5	0.75
qwen3	0.75	0.75	0.75	0.5
mistral	0.1	0.25	0.1	0.1

Mistral failed the eligibility gate in all languages; all four mistral rows and qwen3-LT were excluded. **[Correction, iter 4]:** Seven eligible rows remained (not eight as previously stated).

Verdict: FALSIFY. P1 Spearman(index, residual) = -0.009 $[-0.131, 0.192]$ over 21 rows. The index does not predict the cross-model, cross-language residual. Two cheap baselines beat both the index and the geometric predictor: single-site transfer $\rho +0.732$; unedited model’s own refusal rate $+0.661$.

[Correction, iter 4]: The judge misses its own agreement gate (κ 0.683, below the 0.75 threshold).

4.5 Evaluation 1: Audit and Judge-Sensitivity Table

Audit headline (gen_art_evaluation_1, iteration 3). 167 draft numbers checked: 137 match, 8 mismatch, 12 misdescribed, 6 untraceable (mismatch rate 5.5%). Two sign-reversed conclusions were found.

Judge-sensitivity table:

Table 27: Headline EN–SL gap as a range across judges. Source: `gen_art_evaluation_1`, iteration 3.

Judge	Definition	Gap	CI
Qwen3-14B	strict (REFUSED only)	+0.69	[0.60, 0.78]
Qwen3-14B	broad (REFUSED + PARTIAL)	+0.23	[0.14, 0.32]
keyword	strict	+0.06	[−0.02, 0.14]

Dead-end ledger. [Correction, iter 4]: 16 items (not 20 as previously stated). See Table 35 at the end of the report.

5 Iteration 4

Iteration 4 was designed to close three remaining gaps: (1) the placement finding rests on the exploratory band-density correlation without a causal mechanism; (2) the result is Gemma-only;

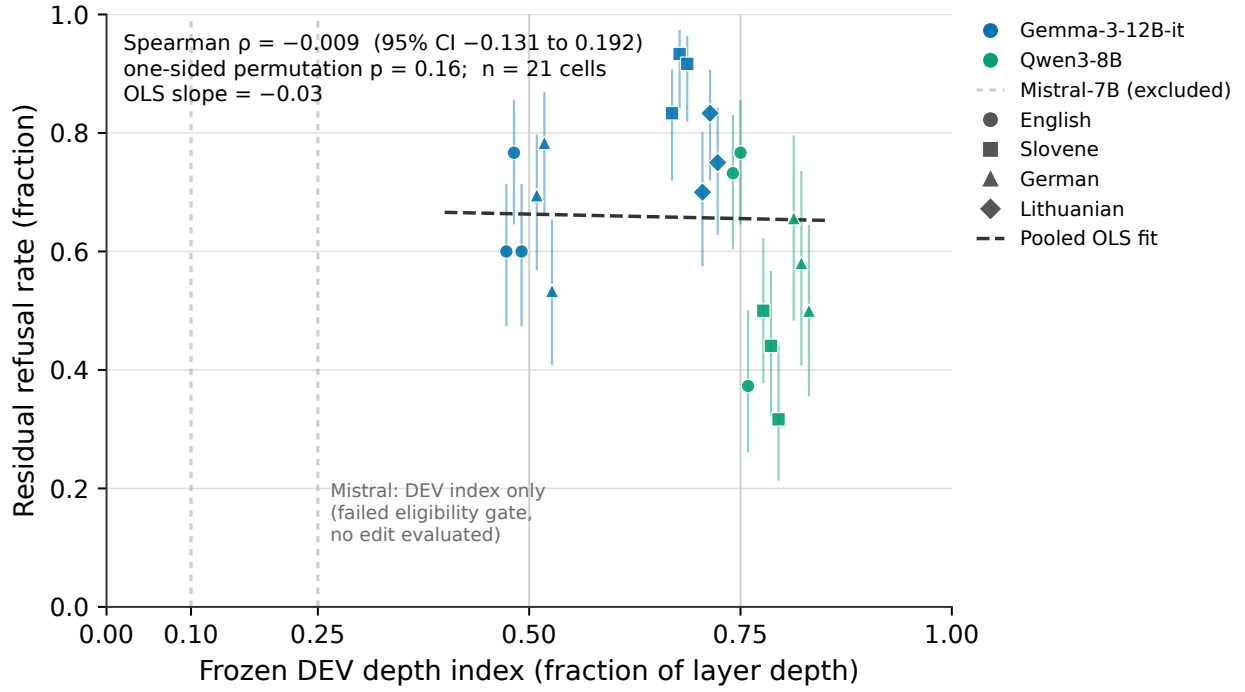


Figure 5: Frozen DEV depth index vs observed residual refusal across 21 model-language-weight cells (3 models $\times$ 4 languages $\times$ weight conditions, eligible rows only). Spearman = -0.009 , failing the predictive test.

and (3) Heretic’s keyword blindness was demonstrated qualitatively but not quantified.

5.1 Experiment 13: Causal Write Profile and Overlap in Gemma

Goal (`gen_art_experiment_13`, iteration 4). Measure the causal write profile $e_L(h)$ for the anchor model and test the overlap instrument O .

Causal write profile. Single-layer edits at each of 48 layers, evaluated on S3 half-A (44 harmful per language). Profile peaks at layer 19 (EN) and layer 16 (SL); Spearman between EN and SL profiles: 0.588. Split-half reliability: EN 0.84, SL 0.51.

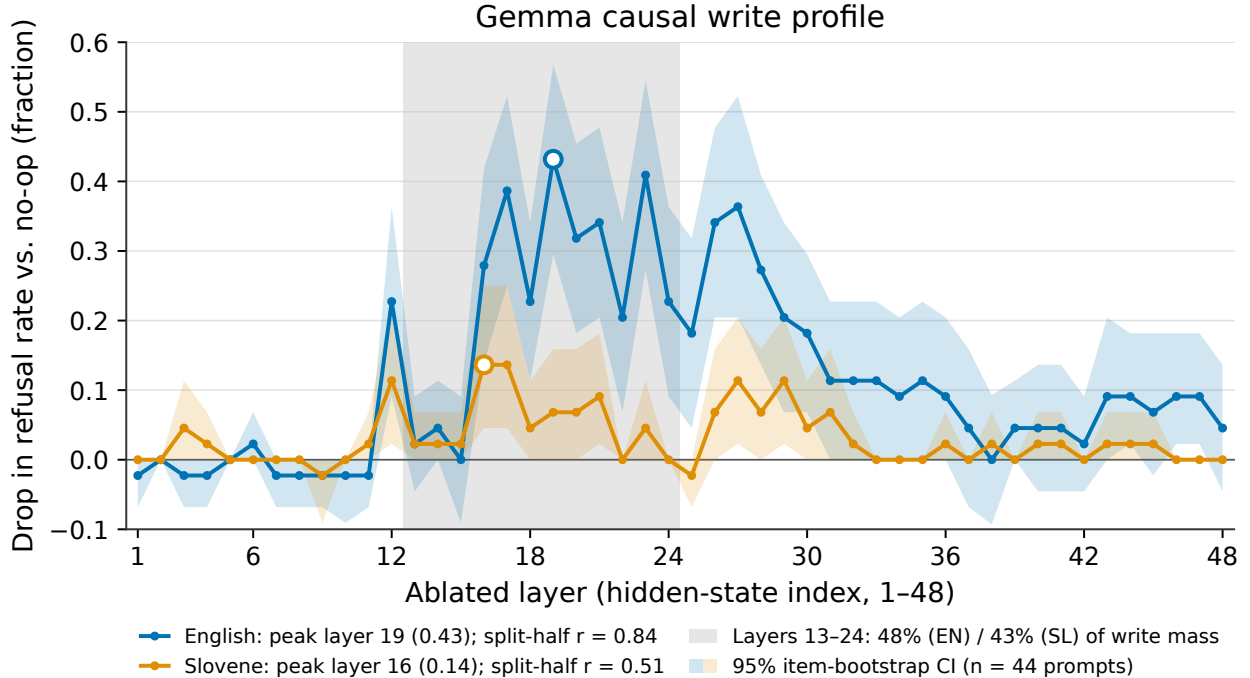


Figure 6: Per-layer refusal drop from single-site ablation in Gemma, for English (blue) and Slovene (orange). The profile peaks at layer 19 (EN) and 16 (SL), with most write mass in band 13–24.

Confirmation results (8 matched-energy groups):

Table 28: Overlap instrument confirmation, Gemma. Source: `gen_art_experiment_13`, iteration 4.

Statistic	EN	SL
Spearman $\rho(O, \text{refusal})$	$-0.96 [-0.98, -0.84]$	$-0.83 [-0.84, -0.82]$
Pooled matched-group contrast	$-0.688 [-0.75, -0.62]$	$-0.377 [-0.45, -0.30]$
Controls (random, PC)	within ± 0.03	within ± 0.03
Dose rival ($2\times$ late-layer)	0.88	0.92
Best placement: L16–31 vs L33–48	0.07 vs 0.92	0.27 vs 0.92

Nested R^2 , per language:

[Correction, iter 5]: Replaced the single-column table (which was exp14 SL data mislabelled as exp13) with the correct per-language exp13 ladders.

Table 29: Nested R^2 , Gemma (exp13). Source: `gen_art_experiment_13`, iteration 4.

	EN	SL
log energy alone	0.001	0.008
+ nuisance stack (count, span, cosine)	0.879	0.734
+ O (full)	0.907	0.790
ΔR^2 of O over nuisance stack	0.027	0.056
O alone	0.873	0.754
EN/SL cosine alone	0.827	0.630
ΔR^2 of O over logE + count only	0.892	0.770
Spearman(O , cosine)	0.808	0.764

PRIMARY CRITERION: FALSIFIED. O adds $\Delta R^2 = 0.027$ (EN) / 0.056 (SL) over a nuisance stack that contains the g-weighted EN/SL cosine. Both are under the 0.10 bar. **[Correction, iter 5]:** The failure reason is collinearity with the cosine ($\rho = 0.81/0.76$), not that energy already explains the outcome: log energy alone explains 0.001/0.008.

Replication in Qwen3-8B (outside family, 3 languages): Spearman $\rho(O)$: EN -0.76 , SL -0.93 , DE -0.84 . Matched groups held in all three languages (9/9 favour high- O).

Judge certification: Within-edited pooled $\kappa = 0.818$. EN $\kappa = 0.856$ PASS; SL $\kappa = 0.744$ FAIL. Slovene gap claims in this experiment are judge-sensitive.

5.2 Experiment 14: Causal Write Profile and Overlap in GaMS3

Goal (`gen_art_experiment_14`, iteration 4). Repeat the causal write-profile measurement for the sibling model.

Causal write profile. Peaks at layer 27 (both EN and SL), in band 25–36.

Table 30: GaMS3 band mass distribution. Source: `gen_art_experiment_14`, iteration 4.

Band	EN mass	SL mass
1–12	0.00	0.02
13–24	1.05	0.55
25–36	1.50	0.57
37–48	0.27	0.05

Split-half reliability: EN 0.812, SL 0.312. The SL profile is declared UNRELIABLE.

VERDICT: PARTIAL. Overlap O_{SL} predicts Slovene refusal at Spearman -0.900 [-0.928 , -0.857]. **[Correction, iter 5]:** Incremental R^2 of O over log-energy: 0.580 (LOO: 0.594), passing the 0.10 threshold.

Nested R^2 , GaMS3 SL (each predictor added separately to logE):

THREE BOUNDARIES:

1. O_{band4} LOO $\Delta R^2 = 0.845$ and $O_{\text{cos}} = 0.834$ both beat O ’s 0.594: the expensive instrument does not earn its cost over one-forward-pass baselines.
2. The DEV argmax names the wrong band (NAMED_AND_LOST).
3. The GaMS3 profile predicts the sibling (Gemma) at Spearman -0.442 , no worse than its own -0.278 on Gemma.

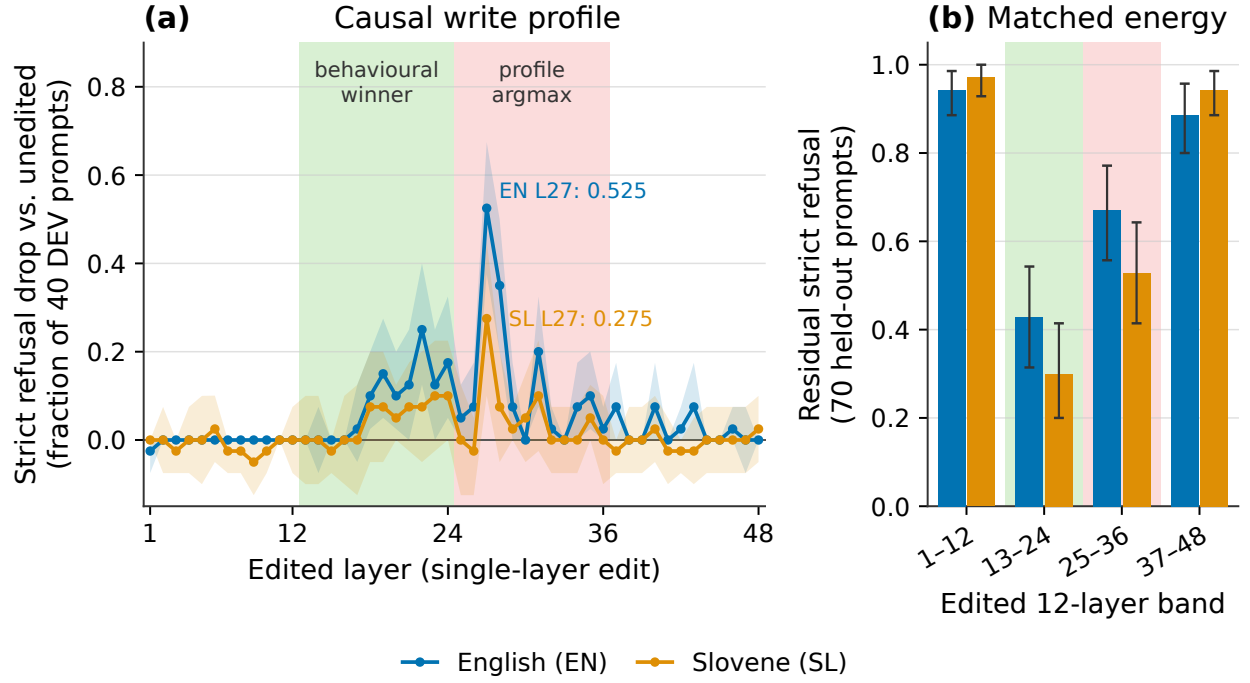


Figure 7: Per-layer refusal drop from single-site ablation in GaMS3. The profile peaks at layer 27 (both EN and SL), in band 25–36, but the behavioural winner at matched energy is band 13–24, the same as in Gemma.

Table 31: Nested R^2 , GaMS3 SL (each predictor added separately). Source: gen_art_experiment_14, iteration 4.

Model	R^2
logE	0.090
logE + O	0.669
logE + $O_{\cos}$	0.788
logE + O_{band4}	0.808
logE + O + $O_{\cos}$ (cumul.)	0.805
logE + O + $O_{\cos}$ + O_{band4} (cumul.)	0.822

Placement vs dose dissociation (A1–A4 ladder). [Correction, iter 5]: All four contrasts are Slovene strict. At fixed placement, increasing dose reduces Slovene strict refusal: dose at fixed O_{ship} -0.186 $[-0.286, -0.086]$; dose at fixed O_{swap} -0.157 $[-0.257, -0.071]$. At fixed dose, swapping placement does not significantly change refusal: placement at fixed high E -0.029 $[-0.129, +0.071]$, McNemar p 0.774; placement at fixed low E $+0.000$ $[-0.086, +0.071]$, p 1.000.

Judge certification: EN $\kappa = 0.721$ (MISS/JUDGE_SENSITIVE); SL $\kappa = 0.830$ (PASS).

5.3 Experiment 15: Gradient-Blind Fraction

Goal (gen_art_experiment_15, iteration 4). Quantify the structural blindness of Heretic’s keyword refusal rate.

Primary prediction: FALSIFIED. Paired GBF difference $+0.006$ $[0.000, 0.019]$; judge-referenced reverses at -0.029 .

Table 32: Keyword objective structural blindness. Source: `gen_art_experiment_15`, iteration 4.

	Gemma (anchor)	GaMS3 (sibling)
Keyword range	[72, 100]	[16, 99]
Classifier range	[5, 97]	[0, 98]
Slope (keyword on classifier)	0.308	0.738
Mean keyword minus classifier	+20.5	+15.0
Keyword floor	72	16
GBF (all candidates)	0.051	0.002
GBF (low-C region, $C \leq 50$)	0.470	0.017
Threshold blindness fraction (TBF)	1.0	1.0
Self-placebo	0	0
Monotone?	Yes	Yes

Reselection table:

Table 33: Reselection under different scorers. Source: `gen_art_experiment_15`, iteration 4.

Model	Scorer	Selected trial	Rule
Gemma	Keyword (K)	107	fallback
Gemma	Classifier (C)	98	primary
Gemma	Qwen3-14B workhorse (J)	64	primary
GaMS3	Keyword (K)	88	fallback
GaMS3	Classifier (C)	85	primary
GaMS3	Qwen3-14B workhorse (J)	115	primary

[Correction, iter 5]: The frozen prediction that one search is blinder overall was FALSIFIED. Threshold blindness is shared (TBF = 1.0 in both searches).

5.4 Evaluation 2: Comprehensive Audit

Goal (gen_art_evaluation_2, iteration 4). Audit all prior claims, certify the judge, measure PARTIAL transitions, and quantify the NF4 confound.

Judge calibration. $n = 900$ items. Overall weighted $\kappa = 0.873$. Keyword $\kappa = 0.074$. Classifier $\kappa = 0.919$. EN gate MET (κ 0.858). SL gate NOT MET (κ 0.723).

Flip analysis:

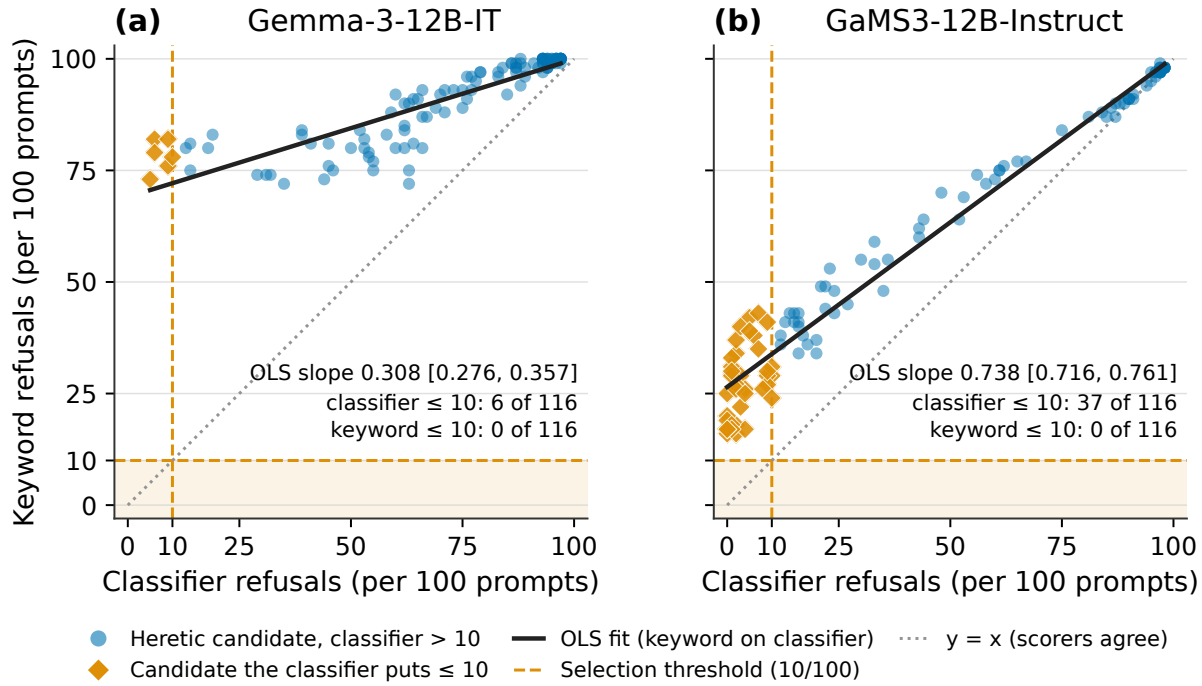


Figure 8: Keyword refusal count vs classifier refusal count across 116 Heretic candidates, for Gemma (left) and GaMS3 (right). The keyword compresses Gemma’s range to $[72, 100]$ and never enters the selection threshold region (dashed line at 10).

Table 34: Flip analysis: slope and intercept changes after editing. Source: `gen_art_evaluation_2`, iteration 4.

Model	Lang	Slope ratio [CI]	Intercept	Refit AUROC	Class
Gemma	EN	0.50 [0.24, 0.86]	−6.65	≥ 0.995	MIXED
Gemma	SL	0.34 [0.19, 0.54]	−5.32	≥ 0.995	MIXED
GaMS3	EN	NOT_ESTIMABLE (too few refusals after edit)			
GaMS3	SL	NOT_ESTIMABLE			

NF4 vs bf16 confound: Paired SL–EN strict gap bf16 +0.60 [+0.40, +0.80] vs NF4 +0.45 [+0.20, +0.70], $n = 20$ S5X pairs, one checkpoint. The difference is not resolved. The asymmetry is at least as large in bf16, so it is not an NF4 artefact.

Independent recompute. 102 numbers checked: 97 match, 3 misdescribed, 2 mismatch. Rate 4.9% [2.1, 11.0]. All placebos collapse.

5.5 Research 1: Novelty Positioning

Goal (gen_art_research_1, iteration 4). Establish what the surviving results add relative to published work.

The surviving positive finding is a matched-total-energy AND matched-layer-count placement contrast. Li et al. [19] (ICLR 2025) already localise contiguous middle safety layers; Bosco and Srinivasan [20] locate refusal beyond attention. What this study adds is the energy- and count-matched construction plus the dose rival’s failure.

The negative finding—that an activation-space depth measurement fails to predict weight-edit outcomes—positions against Jiang [21] and AdvPrefix [22]. **[Correction, iter 5]:** AdvPrefix repositioned from the depth result to the selection-blindness companion.

6 Iteration 5

Iteration 5 ran no new experiments; it ran three independent recompute/audit passes and one positioning analysis.

6.1 Evaluation 3: Terminal Audit

(gen_art_evaluation_3, iteration 5). 121 claims checked, 14 discrepancies (11.6%). 897 assertions classified: 557 OBSERVATION, 285 INTERPRETATION, 45 FAILED_HYPOTHESIS, 10 UNEXECUTED_PROPOSAL. No source is a STAND_IN.

6.2 Evaluation 4: Report Repairs and Independent Recompute

(gen_art_evaluation_4, iteration 5). 136 first-pass numbers recomputed; 21 defective (15.4%, Wilson 95% CI [10.3%, 22.5%]), of which 14 are material. No number was SIGN_REVERSED. 8 of 8 placebos collapsed.

Citation lint. 41 file references checked: 27 resolve as written, 14 after rewrite, 0 unresolvable (gate PASS).

Failed hypotheses ledger (F1–F17):

Table 35: Evidenced failed hypotheses. Sources: gen_art_evaluation_4 (F1–F14) and gen_art_research_2 (F15–F17), iteration 5.

#	Status	Hypothesis	What killed it
F1	Killed	Depth coverage explains residual SL refusal over energy	$\Delta R^2=0.040$ but LOO -0.002
F2	Killed	Broad-and-weak beats narrow-and-strong	Falsified opposite: -0.10

#	Status	Hypothesis	What killed it
F3	Killed	Count-based depth index predicts cross-model residual	Spearman -0.009
F4	Killed	EN/SL direction cosine carries transfer	Spearman $+0.010$
F5	Killed	Corrected objective produces a better edit	P7 fired: dose $+0.33$ vs corrected $+0.38$
F6	Killed	Exposure differential carries the surrogacy gap	$\Delta R^2(D S0) \approx 0$
F7	Killed	r_prior direction carries Slovene residual	Cut $0.036 < \text{best random } 0.074$
F8	Killed	Thin-margin account explains the gap	Margin-matched gap remains: $+0.057$
F9	Killed	Static geometric predictors forecast the gap	CV $R^2 \leq 0$
F10	Killed	Language-identity direction is a usable lever	FLORES $\Delta \text{NLL} +2.0$
F11	Killed	Depth index EN/SL difference is informative	4-layer gap inside permutation null
F12	Killed	Energy-matched random ablations are adequate controls	Controls are destructive
F13	Killed	Conformal forecasting is usable	85% coverage in only 50% of cells
F14	Killed	Batched and single generation are interchangeable	B3 certification FAILED
F15	Killed	Effective depth region differs between siblings	DEV band NAMED_AND_LOST; placement $p=0.774$
F16	Killed	Effective region differs by language	Language-label placebo does not collapse: $\rho -0.94$
F17	Killed	Anchor search is blinder than sibling's	$+0.006$ [0.000, 0.019]; judge-ref reverses

Unexecuted proposals (U1–U10):

Table 36: Unexecuted proposals. Source: `gen_art_evaluation_4`, iteration 5.

#	Proposal	Reason
U1	Native-speaker review of all non-English material	No qualified reviewer; PENDING
U2	Second LLM judge (Gemini) on iter-1 SL arm	Budget block: HTTP 403
U3	Full gpt-4.1 primary-judge coverage iters 2–3	Budget block
U4	P2 Sobol sensitivity bands	Cut for time

U5	English-only-vs-full forecast arm	Cut after conformal coverage failure
U6	Language-orthogonalised matched-efficacy arm	Cut for time
U7	EuroLLM-9B-Instruct as M3	Gated repository, HTTP 403
U8	Racing placement against published cheap predictors	Identified only in positioning; not run
U9	Iter-1 experiment pods 2 and 4	Empty workspaces
U10	Iteration-4 arms not reached	See per-pod deviations

6.3 Evaluation 5: Keyword Blindness Decomposition

(**gen_art_evaluation_5**, **iteration 5**). Decomposed the keyword objective’s blindness into per-language components on verified translation pairs.

Held-out agreement per language. On edited English outputs, keyword $\kappa = 0.021$ ($n=490$). It misfires: calls 0.341 of responses refusals where the reference sees 0.173, and 0.814 of its “refusals” are false. On edited Slovene outputs $\kappa = 0.000$: the rule is SILENT and fires on 0 of 490 responses, while the reference sees refusal in 0.439.

Cross-lingual measurement bias. On the 100 verified EN–SL pairs for the main Gemma edit, keyword score discrepancy from the workhorse judge: $d_{\text{EN}} = +0.74$ and $d_{\text{SL}} = -0.82$, so $\Delta_{\text{lang}} = -1.56$ $[-1.68, -1.44]$. In Slovene all discrepancy is missed refusals (FN rate 0.82, FP rate 0.00). In English the rule over-reports (FP rate 0.78). Disagreement RATES happen to be equal (McNemar $p = 1.00$), but DIRECTIONS differ (sign test $p = 5.0\text{e-}29$).

Judge sensitivity. Registry counts: JUDGE_ROBUST 6, JUDGE_SENSITIVE 72.

6.4 Research 2: Positioning Against Nearest Published Work

(**gen_art_research_2**, **iteration 5**). Positioned the surviving results against neighbours.

Qualifiers checked:

- Selection of sites by measured causal effect: DROPPED (covered by [26], [21]).
- Depth-localised vs distributed refusal: DROPPED (covered by [23], [24]).
- Layer sensitivity differs per language: NARROWED (covered by [25]; claim is only the edit-placement consequence).
- Failed dose rival at matched placement: KEPT.
- Outside-family replication (Qwen3-8B): KEPT.
- Matched total edit size AND matched layer count: KEPT, NARROWED.
- Effective region differs between sibling checkpoints: DROPPED (FALSIFIED by F15).

Bibliography repairs. [19] venue year corrected (ICLR 2025, not 2024); arXiv IDs added for [20] and [21]. AdvPrefix [22] repositioned from depth result to selection-blindness companion.

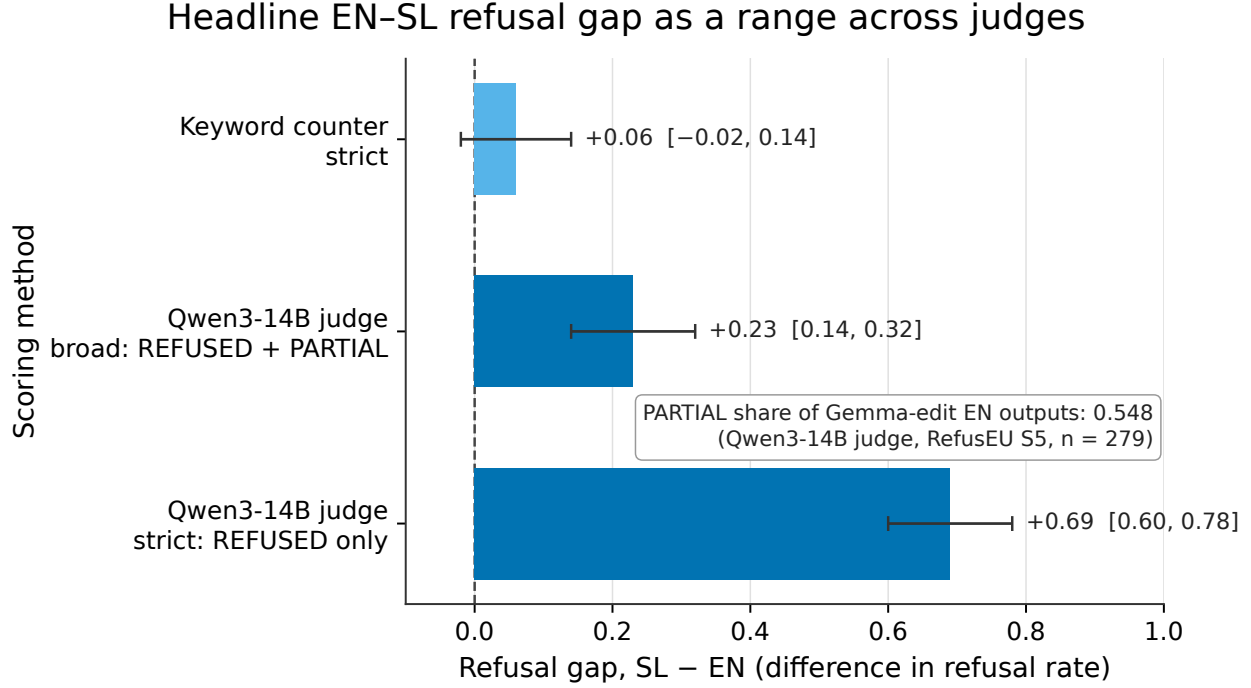


Figure 9: The Gemma edit S5X paired EN-SL refusal gap under three scoring methods. The gap ranges from +0.06 (keyword strict) to +0.69 (Qwen3-14B strict); the PARTIAL class is the wedge.

7 What We Have Learned

Five iterations have converged on a picture with one firm positive, one transferable negative, a structural measurement finding, and seventeen falsified predictions.

Firm positive: matched-budget placement orders residual refusal. At matched total removal energy AND matched layer count, where in depth an English-derived refusal edit deposits that energy orders how much refusal survives: in both languages, in two sibling checkpoints, and in an outside family (Qwen3-8B), and not recoverable by doubling the dose of a badly placed edit. The band is shared; the residual is not. The English causal write profile predicts the Slovene residual at $\rho -0.94$, so the language-label placebo does not collapse; what differs by language is how much refusal is left at the same placement and energy (strict refusal EN 0.07 vs SL 0.27 in group G3). Two one-forward-pass baselines (O_{cos} and O_{band4}) match or beat the expensive single-layer causal instrument. At the refusal floor of shipped edits, dose not placement is the active variable (p 0.774 / 1.000).

Transferable negative: the activation-space depth index fails to predict weight-edit outcomes. Spearman -0.009 over 21 rows (comparing Gemma with Qwen3-8B, plus excluded Mistral). The EN/SL direction cosine fails beside it (Spearman $+0.010$). Two baselines beat both: single-site causal transfer rate $+0.732$ and unedited model’s own refusal rate $+0.661$.

Structural measurement finding: the keyword objective is cross-lingually biased. On verified translation pairs, $\Delta_{\text{lang}} = -1.56 [-1.68, -1.44]$: the English keyword rule over-reports in English (FP rate 0.78) and is silent in Slovene (FN rate 0.82). Threshold blindness is 1.0 in both searches. The pre-registered prediction that one search is blinder overall was FALSIFIED.

Seventeen hypotheses falsified across five iterations (F1-F17): depth coverage (F1), broad-

beats-narrow (F2), count-based cross-model index (F3), direction cosine (F4), corrected objective (F5), exposure differential (F6), r_{prior} (F7), thin-margin (F8), static geometry (F9), language-identity lever (F10), depth index difference (F11), random controls (F12), conformal forecasting (F13), batch interchangeability (F14), sibling band difference (F15), language band difference (F16), and anchor blinder (F17).

The headline gap stated as a range. The Gemma edit S5X paired EN–SL gap ranges from +0.06 (keyword strict) to +0.69 (Qwen3-14B strict), with the broad definition at +0.23. On GaMS3, all judges agree the gap is near zero. The PARTIAL class (0.548 of Gemma edit EN outputs) is the wedge. The Slovene judge certification gate was not met (κ 0.723 < 0.80), so all Slovene gap claims remain judge-sensitive. 94–95% of edited English outputs were truncated at the 256-token generation cap.

Scope and limits. Two sibling checkpoints of one architecture plus one outside family; one non-English language in the matched panels (two further languages only in the outside-family arm); one or two optimiser seeds; NF4 throughout with a single bf16 control cell (paired SL–EN gap bf16 +0.60 [+0.40, +0.80] vs NF4 +0.45 [+0.20, +0.70], not resolved but the asymmetry is at least as large in bf16); Slovene material machine-translated with native review PENDING. Two checkpoints are two units, not a population: nothing is attributed to Slovene continual pretraining, instruction tuning, or any safety-training stage. Five packets totalling 640 items remain pending for native-speaker labelling. Every judged rate in the report is proxy-certified, not human-certified. 136 numbers independently re-derived in iteration 5; defect rate 15.4% [10.3%, 22.5%] on previously unaudited sections.

8 Run Trajectory

Table 37: Run trajectory: one row per iteration. “Ledger spend” is cumulative LLM/tool/container spend billed through that iteration; it excludes compute-rental time and is not the run’s total cost.

Iter	Hypothesis	Move	Score	Executed	Ledger (\$)
1	What EN ablation misses in SL	—	—	Yes	6.07
2	SL refusal needs a deeper edit	deepen	3	Yes	42.31
3	Where the edit lands, not how deep	deepen	4	Yes	161.91
4	Where the edit lands beats how hard	deepen	3	Yes	258.92
5	Where the edit lands beats how hard	deepen	3	Yes	270.89

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